

EXERCISE-6

AIM: Online Shopping menu application using **Grid Layout**

INTRODUCTION TO LAYOUTS

A layout refers to the arrangement of visual elements such as text, images, menus, and buttons on a webpage. It defines the overall structure of a website and helps present content in an organized and user-friendly manner. Proper layout design improves readability, navigation, and user experience. Modern web design uses CSS techniques to create flexible and responsive layouts for different screen sizes.

Types of Layouts

1. Fixed Layout

In a fixed layout, the width of the webpage is set to a specific size. The layout does not change with screen size and looks the same on all devices.

2. Fluid (Liquid) Layout

A fluid layout uses percentage-based widths. The content adjusts automatically according to the screen size.

3. Responsive Layout

Responsive layouts adapt to different screen sizes using CSS media queries, Flexbox, or Grid. It provides a better experience on mobiles, tablets, and desktops.

4. Adaptive Layout

Adaptive layouts use predefined screen sizes and load different layouts based on the device.

5. Grid Layout

Grid layout divides the page into rows and columns. It provides precise control over placement and alignment of elements.

6. Flexbox Layout

Flexbox is a one-dimensional layout system used to arrange items in rows or columns efficiently.

Theory About Online Shopping menu

1. LinearLayout

A layout that arranges UI components in a single direction, either vertically or horizontally.

2. GridLayout

Arranges components in rows and columns, ideal for displaying product items in an online shopping menu.

3. **TextView**

Used to display text such as product name, price, and description.

4. **ImageView**

Displays product images in the shopping menu.

5. **Button**

Represents clickable actions like **Add to Cart** or **Buy Now**.

6. **RecyclerView**

Efficiently displays a large list or grid of products by reusing views.

7. **ScrollView**

Allows scrolling when the content exceeds the screen size.

8. **android:layout_width**

Defines the width of a UI component.

9. **android:layout_height**

Defines the height of a UI component.

10. **android:text**

Sets the text displayed in a TextView or Button.

11. **android:src**

Specifies the image source for an ImageView.

12. **android:id**

Assigns a unique identifier to a UI component for reference in Java/Kotlin code.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:rowCount="4"
    android:columnCount="3"
    android:padding="8dp"
    android:background="#FFFFFF">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="A24126510100"
        android:textSize="16sp"
        android:textColor="#000"
        android:gravity="center"
```

```
        android:layout_columnSpan="3" />
<ImageView
    android:layout_width="150dp"
    android:layout_height="200dp"
    android:src="@drawable/rajii"
    android:layout_gravity="center"
    android:contentDescription="profile"
    android:layout_columnSpan="3" />
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="dairyMilk"
    android:textSize="36sp"
    android:textColor="#000"
    android:gravity="center"
    android:layout_columnSpan="3" />
<LinearLayout
    android:layout_width="0dp"    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_columnWeight="1"
    android:padding="8dp">
    <ImageView
        android:layout_width="280dp"
        android:layout_height="150d"    android:src="@drawable/chocolate_2" />
    <TextView
        android:text="₹499"    android:textSize="14sp"
        android:textColor="#000"
        android:layout_width="match_parent"    android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btn_add_to_cart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add to Cart"
        android:backgroundTint="#FF9800"
        android:textColor="#fff"
        android:padding="8dp"/>
</LinearLayout>
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
```

```
android:layout_columnWeight="1"
android:padding="8dp">
<ImageView
    android:layout_width="280dp"
    android:layout_height="150dp"
    android:src="@drawable/chocolate_3" />
<TextView
    android:text="₹699"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="14sp"
    android:textColor="#000" />
<Button
    android:id="@+id/btn_add_to_cart"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add to Cart"
    android:backgroundTint="#FF9800"
    android:textColor="#fff"
    android:padding="8dp"/>
</LinearLayout>
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_columnWeight="1"
    android:padding="8dp">
    <ImageView
        android:layout_width="280dp"
        android:layout_height="150dp"
        android:src="@drawable/chocolate_4" />
    <TextView
        android:text="₹899"
        android:textSize="14sp"
        android:textColor="#000"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btn_add_to_cart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add to Cart"
```

```
        android:backgroundTint="#FF9800"
        android:textColor="#fff"
        android:padding="8dp"/>
</LinearLayout>
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_columnWeight="1"
    android:padding="8dp">
    <ImageView
        android:layout_width="280dp"
        android:layout_height="150dp"
        android:src="@drawable/chocolate_1" />
    <TextView
        android:text="₹1299"
        android:textSize="14sp"
        android:textColor="#000"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btn_add_to_cart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add to Cart"
        android:backgroundTint="#FF9800"
        android:textColor="#fff"
        android:padding="8dp"/>
</LinearLayout>
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_columnWeight="1"
    android:padding="8dp">
    <ImageView
        android:layout_width="280dp"
        android:layout_height="150dp"
        android:src="@drawable/chocolate_5" />
    <TextView
        android:text="₹599"
```

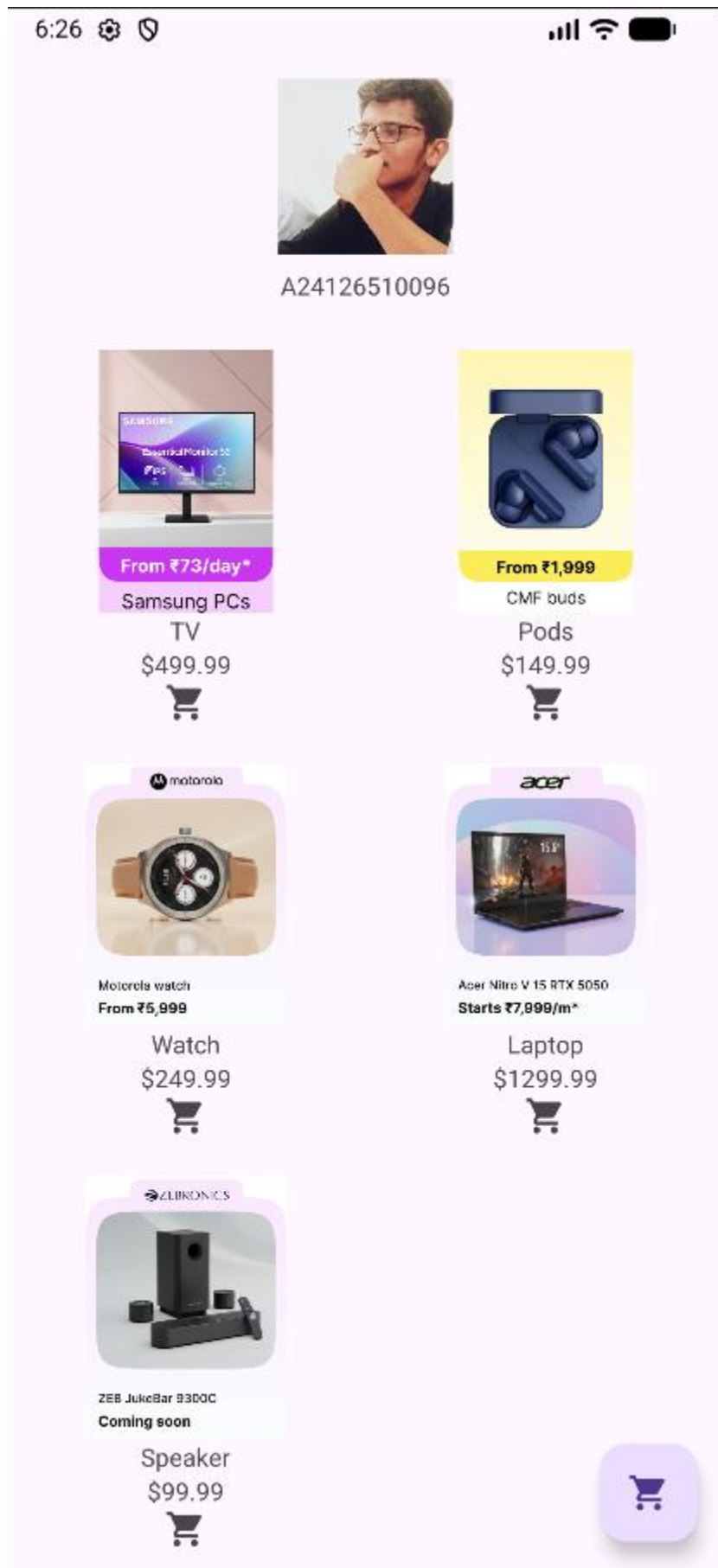
```
        android:textSize="14sp"
        android:textColor="#000"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
<Button
    android:id="@+id/btn_add_to_cart"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add to Cart"
    android:backgroundTint="#FF9800"
    android:textColor="#fff"
    android:padding="8dp"/>
</LinearLayout>
<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_columnWeight="1"
    android:padding="8dp">
    <ImageView
        android:layout_width="280dp"
        android:layout_height="150dp"
        android:src="@drawable/chocolate_6" />
    <TextView
        android:text="₹799"    android:textSize="14sp"
        android:textColor="#000"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btn_add_to_cart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add to Cart"
        android:backgroundTint="#FF9800"
        android:textColor="#fff"
        android:padding="8dp"/>
</LinearLayout>
</GridLayout>
```

Main activity.java

```
package com.example.exce6;
import android.os.Bundle;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

OutPut :



ANIL NEERUKONDA INSTITUTION OF TECHNOLOGY AND SCIENCES
Department of Computer science and Engineering
Mobile Application and Development